

AIx speed: Playback Speed Optimization using Listening Comprehension of Speech Recognition Models

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ABSTRACT

In recent years, more and more time has been spent watching videos for online seminars, lectures, and entertainment. In order to improve time efficiency, people often adjust the playback speed to a speed that suits them best. However, it is troublesome to adjust the optimal speed for each video and even more challenging to change and adjust the speed for each speaker within a single video. Therefore, we propose "AIx speed," a system that maximizes the playback speed within the range where the speech recognition model can recognize and flexibly adjusts the playback speed for the entire video. This system makes it possible to set a flexible playback speed that balances playback time and content comprehension, compared to fixing the playback speed for the entire video.

CCS CONCEPTS

• **Human-centered computing** → *Auditory feedback.*

KEYWORDS

Video Information, Playback Speed, Deep Neural Network, Speech Recognition, Self-supervised Learning,

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1 INTRODUCTION

With the proliferation of video streaming services, there are more and more opportunities to watch videos for various reasons, such as information gathering, study, and entertainment. When watching videos, people often increase the playback speed of videos to be exposed to more videos or complete watching videos faster. Existing research has shown that when watching videos for study, under certain conditions, differences in video playback speed do not affect learning effectiveness and may even improve grades [10, 12]. It has also been reported that 30%–80% of users prefer to watch dramas at high speed, although there are differences among countries [4].

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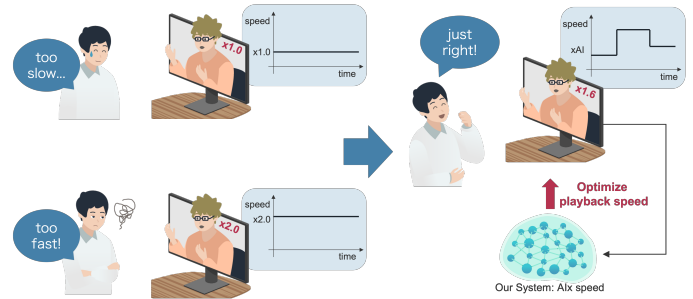


Figure 1: AIx speed flexibly optimizes video playback speed on a video-by-video and within a video. Users can watch videos at a comfortable speed without adjusting the playback speed.

Because of the many benefits of adjusting video playback speed, many methods have been proposed that automatically adjust playback speed based on content and user. The basic strategy of these methods is to speed up the unnecessary parts of the content and delay the unnecessary parts by reading the user's intentions and actions. However, these methods do not focus on the important factor of the playback speed, which is the high or low intelligibility of the conversation in the video. Therefore, users may feel redundant even in the parts of the content deemed necessary or recognized as important by the user. Then, we propose AIx speed, a system that flexibly optimizes the playback speed within a video by increasing the playback speed for conversations with high intelligibility and decreasing the playback speed for conversations with low intelligibility, as shown in Figure 1. The system is a neural network-based model that maximizes both speech recognition performance and video playback speed simultaneously, utilizing the listening comprehension of the speech recognition model, which is said to be comparable to human performance [18]. The playback speed is expected to be slower in areas of the model where speech recognition is complex and faster in areas where speech recognition is accessible, i.e., redundant parts. Additionally, we have conducted preliminary experiments to evaluate the performance of AIx speed. The results show that our system can convert videos with a good balance between playback speed and listening comprehension.

2 RELATED WORK

There are two main methods for adjusting video playback speed to improve the time efficiency of video viewing. One method focuses on the content and removes unnecessary parts; the other method picks up the parts necessary to the user through user interaction.

The former removes portions that do not have audio or subtitles, such as CinemaGazer [9], or portions of sports games that are not highlights of the game [7]. The latter has been studied extensively, mainly in the field of HCI, and adjusts the playback speed based on the user’s behavior. For example, SmartPlayer [3] learns the optimal playback speed based on the user’s past viewing behavior history. There is also technology that allows the user to make rough selections in advance of what he or she considers important and then fast-forwards the rest of the video [5, 8]. Others monitor the user’s movements and adjust the playback speed according to the user’s level of concentration [17] and comprehension [13, 19]. Although these techniques have the advantage of seeking the optimal playback speed for each user, they cannot reflect the effects of essential factors such as the high/low intelligibility of the conversation in the video and its transitions in the playback speed.

3 AIX SPEED

This system consists of two mechanisms, as shown in Figure. 2. One is a playback speed adjuster (left), and the other is a speech recognizer (right). The playback speed regulator is a linear layer attached to a self-supervised learning model using audio data such as HuBERT [6] or Wav2Vec [1, 2, 16] as input. This mechanism is trained to maximize the playback speed with video audio input. The speech recognizer also consists of a linear layer joined to a self-supervised learning model. It takes as input the video audio whose playback speed is varied by the playback speed adjuster and learns to maximize the recognition performance of the input audio. Thus, the error function is as follows:

$$Loss = CTC\ Loss + \lambda \left(\frac{1}{10} \right)^{Avg.\ speed}$$

Learning to minimize this error function calculates a playback speed that balances playback time and content comprehension.

4 PRELIMINARY STUDY

4.1 Implementation

We have created a prototype of AIXSpeed for English speakers. The self-supervised learning model for the playback speed regulator and speech recognizer was the pre-trained Wav2vec2.0 [2] published on fairseq [14], with an optimization function of AdamW [11], an initial learning rate of 10^{-4} , and a batch size of 32. The final linear layers are all 768-dimensional MLPs, and the parameter λ of the error function is set to 10^{-7} . The training datasets were the train-clean-100 and dev-clean sets from LibriSpeech [15].

4.2 Setting

To demonstrate that the proposed method can optimize the playback speed while maintaining content comprehension, we compared the performance of the proposed method with that of the baseline using test-clean sets from LibriSpeech. Performance was evaluated in terms of average playback speed and speech recognition performance (CER, WER). The baseline was the same as our speech recognition model, wav2vec2, with the input speech speed increased by 1x and 2x.

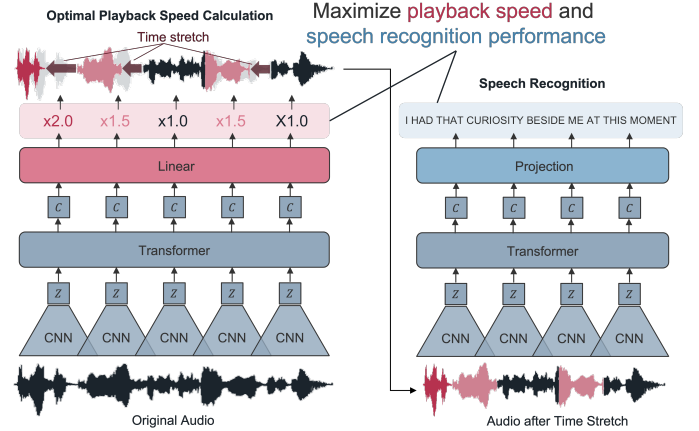


Figure 2: AIX speed architecture: Our system simultaneously optimizes the playback speed regulator (left) and the speech recognizer (right). By doing so, we can maximize the playback speed to the extent that the model can recognize.

Table 1: Performance Comparison of Models

Model	Avg. Speed	CER	WER
Wav2Vec2 (x1.0)	1.00	2.25	5.86
Wav2Vec2 (x2.0)	2.00	68.12	84.67
AIX speed	1.80	6.05	14.42

4.3 Results

Table 1 shows the performance of the model. It can be seen that the proposed model is in between the two baselines for both playback speed and speech recognition performance. In particular, compared to the 2.0x baseline, the proposed model can reduce recognition errors by more than 80%, even though the average playback speed is 1.8x and close to 2.0x. In light of these results, it can be said that the proposed model can maximize playback speed while guaranteeing recognition performance.

5 APPLICATIONS

Examples of AIX speed applications are shown in Figure 3. The first is an example of watching a lecture, movie, or class faster. There are two speakers in the video, and the optimal playback speed can be set for each speaker. The second is an example of speeding up the speech of a non-native speaker to make it easier to listen. Non-native speakers’ speech is often slower than native speakers’ speech, and by speeding it up appropriately, it becomes easier to listen to. The change in playback speed shows that the overall playback speed is faster than the speech between native speakers in the first example.

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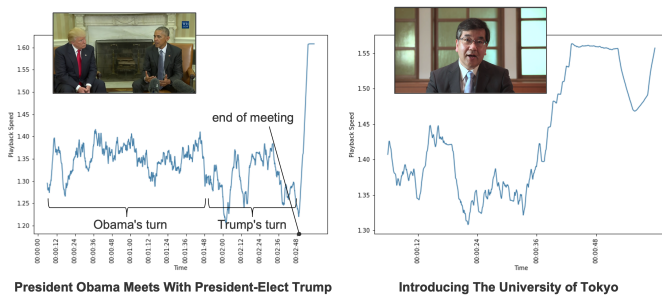


Figure 3: Application Examples

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